



Social4Fashion: An intelligent expert system for forecasting fashion trends from social media contents

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ABSTRACT

The fashion field is continually expanding and evolving, and social media play a significant role in shaping current fashion trends through the influence of online personalities, such as influencers. As a result, fashion designers often turn to social media to gain insights into the latest trends and draw inspiration, while in the past they used to physically visit fashion districts. To automate and speed up this process, an expert system is much needed; thus, Social4Fashion has been created, an end-to-end framework that leverages deep learning-based techniques in order to support creatives in their research and decision-making process, with the final goal of analyzing and predicting trends. This system employs several steps, starting with the automatic data collection from Instagram, using hashtags provided by domain experts. Next, retrieved images are filtered to remove non-fashion related pictures, leaving only those pertaining to the fashion area for further processing. Then, to obtain more specific information about the images, the handbags present (if any) are detected and classified, based on their type; finally, dominant colors of the handbags are retrieved through clustering on the images. All the data collected with this system are then stored and analyzed via user-friendly dashboards, created with the objective of highlighting relevant information, in order to perform analysis on current and future fashion trends. Results show the effectiveness of the proposed system, with an accuracy of 97% (95% confidence interval 0.95–1) for the fashion image classification and a mAP of 0.77 (95% confidence interval 0.73–0.82) for the handbag detection, which makes it suitable for fashion domain analysis. Also, as a result of this work, a novel fashion-related dataset has been made available to the research community. This system can greatly improve the way fashion trends are analyzed, and allow for more efficient and effective design processes in the future.

1. Introduction

According to Statista¹, after difficulties in 2020, during the Covid-19 pandemic, when sales across fashion industry took a slowdown, the global demand for clothing, bag and shoes is set to rise again. The revenue of the global apparel market amounted to about 1.5 trillion U.S. dollars in 2021 and was predicted to increase to approximately 2 trillion dollars by 2026. Fashion knowledge plays a decisive role in this domain, since it directs not only physiological users' needs, but also the requests of social activities and conventions (Ma et al., 2019). Fashion analysis and forecasting is a difficult and complex task but pivotal for the growth of the fashion sector. Fashion trends analysis are gaining increasing importance in the fashion domain as it

offers information about upcoming and outgoing styles. It is a starting point in the product development process (Wazarkar & Keshavamurthy, 2020). Fashion trends are time-dependent. They frequently change due to the effect of world events, social and cultural activities, economic circumstances, technical progresses, charismatic personalities in the society and fashion leaders like fashion designer and celebrity icons.

The advent of social media evolved fashion industry since influencers dictate trends rather than brands (Mameli et al., 2021). Social media persuaded consumers to adopt and move on from fashion trends quicker than before, and, to satisfy customers, brands must sustain with forecasts. Forecasting fashion trends by analyzing social media images allows fashion brands to unlock their potential for commercial and

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creative success. Each social media channel has several insights to offer, giving new ways to achieve goals. Facebook, Instagram, Pinterest, TikTok, Twitter, Flickr, and YouTube play an important role in inspiring paradigm fashion shift as millennials and Gen Z-ers have become both influencers and influencees (Zhou et al., 2021). Social networking sites are the most prevalent sources of data, which are available in several forms like numeric (e.g., number of likes or tag), textual (e.g., captions, hashtags or comments), images (e.g., profile/timeline picture, posts), audio (e.g. songs), video (e.g. funny videos, trailers). Multimedia contents as images are one of the most expressive kind of data because social media pictures contain interesting and useful information.

Along with creativity and intuition social media data get huge importance as decision support systems or creativity support systems, being helpful to achieve information about attributes influencing fashion trends. Analyzing social media pictures is very challenging due to the inner nature of these data, as they are voluminous, unlabeled and heterogeneous. Analyzing images from social media can give useful insights on new trends, but getting data in a manual way is a time-consuming process, having to filter relevant content, extract the information needed to finally analyze what has been found. The adoption of Artificial Intelligence (AI) approaches can ease these steps, providing an automatic method to get relevant images and extract useful information from them. Several AI tasks such as classification, clustering, object detection are valuable for automatically processing these data.

In this context, this work proposes Social4Fashion, an expert system designed for collecting data from Instagram and leveraging Machine Learning (ML) and Deep Learning (DL) algorithms to:

- Define if an image contains one or more fashion item;
- Extract the garment found and discover its type;
- Get the dominant color of the garment.

Today, fashion professionals manually hunt for fashion-related posts on social networks in major cities and locations, taking up a considerable amount of time. Social4fashion seeks to revolutionize this method by automating the trend analysis and comparison process. This will enable the forecasting of future trends and support fashion experts in creating collections that cater to the specific tastes and demands of customers in different areas.

Social4Fashion can be used by companies and designers to discover new trends in fashion and assist them in the design process. It provides an easy-to-use, visual dashboard to monitor customers, to capture trends in communities and, by iteratively applying multidimensional data analysis and filtering, to deeply analyze posts and influencers. Social4Fashion aggregates data from social source and performs analysis via a specifically designed AI pipeline. This work focuses on handbags analysis, but the system can be easily extended to other categories of garments, e.g. shoes, dresses, shirts.

The main concept of Social4fashion is schematically depicted in Fig. 1, which highlights the important contributions of this research.

In summary, the main contributions of this paper, which aims to fulfill the gap in fashion trends forecasting, are (i) the development and deployment of an expert system which leverages AI for augmented creativity, (ii) the development of a social media crawler that scrapes posts from Instagram, (iii) the training and integration of Deep Neural Networks (DNNs) models, which will serve for classification of fashion images, to identify bags in images and then to extract the dominant color of fashion items; (iv) a real-environment test to demonstrate the effectiveness of our approach (v) the collection and release of a new fashion related dataset, for fashion/non-fashion binary classification.

The rest of the paper is organized as follows: Section 2 lists different relevant related works in the fashion field. Section 3 discusses the methodology used. Section 4 discusses experiments and results obtained. In Section 5 some analysis that the system allows to create are presented. Finally, Section 6 contains conclusions and possible future works.

2. Related works

Social media play a crucial role in assessing current fashion trends; with millions of users sharing their personal style and showcasing the latest fashion finds, social media platforms like Instagram, TikTok and Pinterest provide a wealth of information on what is in and what is out in the world of fashion. This constant stream of up-to-date content allows individuals and businesses to stay informed and adapt to the changing trends in real-time. This is why out of the numerous studies on fashion-related issues, our focus is on those that pertain to the fashion trend analysis. AI models can now support industry experts and customers in their decisions. Fashion domain has undergone analysis of various appearance-based tasks, ranging from clothing detection (Kim et al., 2021), classification (Jain & Kumar, 2020; Kolisnik et al., 2021; Seo & shik Shin, 2019; Wang et al., 2018), and generative models for virtual try-on Li et al. (2021). Fashion image generation was also tackled by Rostamzadeh et al. (2018) who introduced a new fashion themed text-to-image generation dataset, with high-quality images and extensive annotations provided by fashion experts.

Several works focus on the representative problem of learning compatible clothing style using deep learning models for style matching. Kiapour et al. (2015), addressed the problem of matching a real-world example of a garment item to the same item in an online shop. Veit et al. (2015) presented a new learning framework that can recover a style space for clothing items from co-occurrence information as well as category labels, while Lin et al. (2019) used a new approach for outfit complementary item retrieval, designing a category-based subspace attention network that takes a single image and two category vectors to form an attention mechanism.

A significant aspect of fashion is its constant evolution and creation of new trends. Fashion companies are, nowadays, more and more focused on predicting trends in terms of garments and their predominant colors in order to create new collections in line with customers' tastes of a specific geographic area. Mall et al. (2019) proposed a framework for automatically analyzing temporal trends in fashion attributes and style by examining millions of photos published publicly to the web. Trends has been characterized using a new model that is both more interpretable and makes better long-term forecasts with respect to the state-of-the-art. The authors recognized however the need for further investigations such as the identification and mitigation of biases in social media imagery, and the propagation of styles across space. Al-Halah et al. (2017) developed a vision-based fashion forecasting model that discovers fine-grained visual styles from large scale fashion data in an unsupervised manner. The model identifies unique style signatures and provides a semantic description for each based on key visual attributes. Furthermore, based on user consumption behavior, the model predicts the future popularity of the styles, and reveals their life cycle and status (e.g. in or out of fashion). Ma et al. (2020) proposed KERN (Knowledge Enhanced Recurrent Network) framework in which they developed a multi-horizon forecasting by incorporating both time series inputs and the associated sequence information collecting a dataset from social media, with the goal to uncover the intricate patterns in the historical time-series records of fashion elements and thus predict future trends. The work by Ding et al. (2022) also addressed the fashion trend forecasting problem based on social media, over 170 fashion elements were studied and three types of user information were involved towards forecasting insightful fashion trends for specific user groups. They proposed a model called Relation Enhanced Attention Recurrent network (REAR) to utilize multiple relationships between specific fashion trends to capture the intricate patterns in time-series data and accurately predict future fashion trends.

Fashion domain is already benefiting from expert systems, one notable example is the novel proposal by Shimizu et al. (2023) which introduced a research area called "fashion intelligence" and a fashion intelligence system that enables the interpretation of abstract fashion attribute information based on a visual-semantic embedding method.

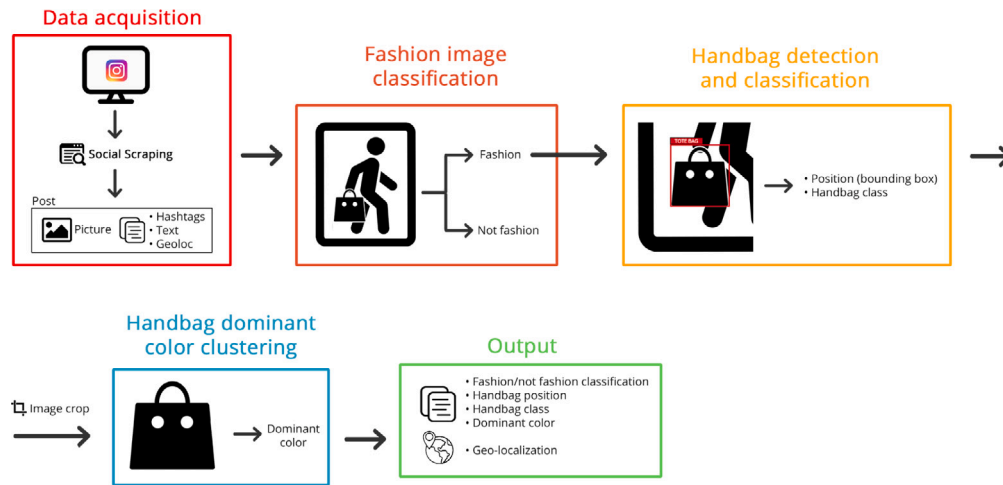


Fig. 1. Workflow of the proposed system. The first step involves the acquisition of data from Instagram posts; then, there is the fashion/non-fashion classification of the picture; the third step involves the detection of the position and the class of the handbag; in the fourth step the dominant color is found. Finally, a result is produced, containing the new acquired information.

The proposed system utilizes methods such as image retrieval by calculation of rich tags and full-body outfit image similarities and visual interpretation support by an attribute activation map. The effectiveness of the proposed system was demonstrated by applying it to real-world service datasets and conducting evaluation experiments.

In fashion trends analysis, a pivotal role is color analysis and forecasting. Color forecasting deals with highly unpredictable and volatile time series such as the fashion color trend, [Yu et al. \(2012\)](#) in this field investigated various forecasting models. Statistical model ARIMA (Autoregressive Integrated Moving Average), the GM (Grey method) family models (GM, GNNM — Grey neural network model, and Improved GM), the ANN (Artificial neural network) family models (ANN, ELM — Extreme Learning Machine), and a new method called GRA-ELM were employed in the analysis of the fashion color trend forecasting. The proposed hybrid method GRA-ELM achieved very good forecasting accuracy and also avoided the time consuming drawback in ANN. Color palette extraction from fashion runway images was tackled by [Lai and Westland \(2020\)](#) using a classical machine learning approach, K-Means clustering. Extracting dominant colors using k-means algorithms has been proven effective also by [Han et al. \(2022\)](#), who tried to answer the question whether fashion designers apply these colors in fashion shows that guide seasonal fashion trends. The study analyzed image data from fashion collections through machine learning to obtain measurable results by web-scraping catwalk images, separating body and clothing elements via machine learning, defining a selection of color chips.

Finally, expert systems are emerging in fashion domain, especially for outfit recommendation, [Lu et al. \(2019\)](#) utilized the hashing technique for efficient personalized fashion outfit recommendation first, then [Lu et al. \(2021\)](#) proposed a learnable personalized anchor embedding (LPAE) approach for personalized outfit recommendation and new user profiling. The model encode the outfit into a compact embedding using the self-attention mechanism to capture the high-order relationships among fashion items. [Li et al. \(2017\)](#) instead considered the challenging problem of fashion outfit composition, which reflects the difficulties of matching domain expert knowledge and modeling the diversity in fashion. They proposed an outfit quality scorer which is an end-to-end trainable system, achieving promising performance. [Balim and Özkan \(2023\)](#) proposed a model to predict and diagnose outfit compatibility using a semantic segmentation technique and encoder-decoder models to predict and diagnose it. [Song et al. \(2017\)](#) presented a content-based neural scheme (BPRDAE) for compatibility modeling towards clothing matching (i.e., matching tops and bottoms), which is able to jointly model the coherent relation between different modalities of fashion items and the implicit preference among items via a dual

auto-encoder network. [Iwata et al. \(2011\)](#) developed a recommender system for clothing coordinates using full-body photographs from fashion magazines. The idea is to apply this method also when purchasing clothes that match one's own clothes. Even online retailers can benefit from these models in order to propose additional garments to match with the item already purchased by the customer. Other approaches in the fashion field implement models to recommend visually compatible products based on scene images ([Kang et al., 2018](#)) or based on the general context ([Cucurull et al., 2019](#)) of the image itself. [Wu et al. \(2019\)](#), instead, proposed a work to investigate natural language user feedback and attributes for dialog-based image retrieval. In this way, the preferences and choices of the user affect the output of the model in order to obtain a result as close as possible to the needs of the customer.

To the best of our knowledge there is still a gap in the fashion domain experts system, which is related to the possibility of leveraging social networks as a continuous source of data for fashion trends analysis, in order to provide fashion-domain expert with a useful tool to analyze current geo-localized fashion trends.

Social4fashion hence aims to create an expert system to analyze fashion trends in the handbags model field by extracting fashion data from Instagram. This represent a novel completely modular pipeline that paves the road for future development in this field with analysis on current data by also taking into account the localization of a specific trend.

3. Methodology

The proposed system consists of a novel framework, which aims to leverage effective Deep Learning techniques to automate different tasks (that would have had to be done manually by an expert otherwise), with the final goal of generating automatic analysis of relevant data taken from Instagram and, by collecting said data, predict future trends in the fashion field and assist creatives and designers in their decision-making process. This framework poses itself as a novel standard for fashion images classification and handbags detection.

The system is composed of multiple subsequent phases: in the first step, images and information coming from Instagram posts are captured and stored using a crawling algorithm that has been created for this purpose. After data storage, a filtering phase follows, using a VGG-16 ([Simonyan & Zisserman, 2014](#)), with the aim of checking whether fashion elements are present or not inside the picture. If the image has been classified as a fashion image, YOLOv5 ([Jocher, 2020](#)) is applied, in order to detect and localize the exact position of the bag inside the starting image and classify it, based on the type of handbag. In

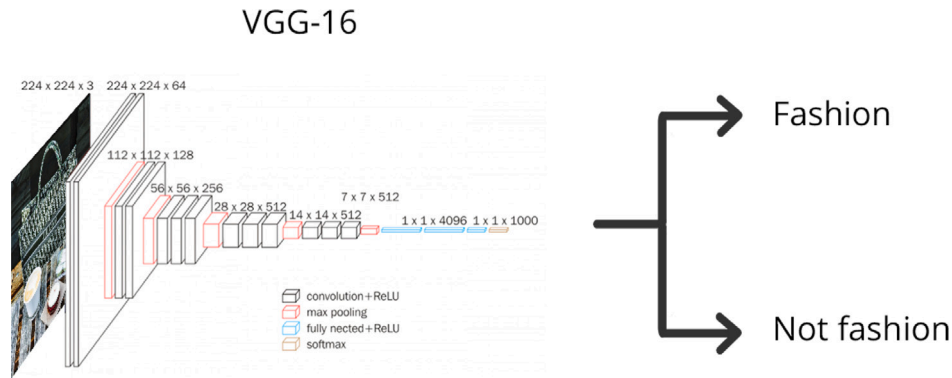


Fig. 2. Workflow of the fashion/non-fashion classification phase with VGG-16.

this way, the detected handbag can be cropped and a new image is obtained with the extracted bag in foreground, ready to be used for subsequent analysis. In fact, through the implementation of K-means clustering (Jain & Dubes, 1988), the dominant color have been detected to evaluate the most popular handbag colors for a specific time period and in a specific geographic area. While fashion trends can evolve rapidly, they also exhibit cyclic patterns over time (Cho, 2019). Instagram, as a platform, provides a unique advantage by offering not only a snapshot of current trends but also a repository of historical images. This inclusion of both recent and older pictures allows for a comprehensive understanding of how fashion trends have evolved, resurfaced, and shaped the industry's landscape over time. Given this assumption, we believe that the validity of the fashion/non-fashion classification has the potential to persist over an extended period, while the bag detection module could be incrementally re-trained to identify new types of bags as they emerge in the market. The modularity of the system facilitates this task conveniently. A graphical representation of these steps is reported in Fig. 1 and are detailed in the next subsections.

3.1. Data extraction and storage

This step represents the first phase of the entire process and could be considered the most delicate one. In fact, since the final objective is to analyze and exploit current trends, it is essential to take into account real and above all up to date data. In order to comply to these factors, a crawling algorithm on Instagram was implemented and executed twice a day. The crawler uses the Instaloader² library to connect and download posts from Instagram. Data are collected by searching hashtags of specific handbag models, defined by domain experts (e.g. shoulder bags, clutch, backpack, etc.) and downloading only information relevant to the aim of the work such as: the picture of the post, post description and hashtags, user information and geo-localization (if present). The downloaded images are stored in a AWS (Amazon Web Services) S3 (Simple Storage Service) bucket, while the text data is inserted in a JSON file (which holds a reference to the image) and stored in a document-oriented database, namely MongoDB.³ The main reason for the extraction of this additional data is to analyze fashion trends from a geographical and temporal point of view, as it is essential to take into account not only the image itself, but also the data accompanying it.

Once this phase ended, a filtering step followed.

3.2. Fashion/non-fashion image classification

One of the main characteristics of the social network is the huge freedom of expression. In this specific case, people are allowed to

publish their photo and write any description they want. We also acknowledge the important role that hashtags play as metadata accompanying images on social media platforms. However, it is crucial to emphasize that the use of hashtags presents complexities stemming from the potential inconsistency between their intended meaning and the visual content they accompany. While we do consider hashtags as part of our data collection process, we exercise caution in relying solely on them for trend analysis. The reason for this cautious approach is twofold. First, hashtags can be employed in a non-coherent manner, leading to situations where their intended significance may not align with the actual content of the images they are paired with. This disparity can result in misinterpretations and misleading conclusions if not appropriately addressed. Secondly, the classification of images into the categories of fashion and non-fashion forms a foundational step in our framework's analytical pipeline. By considering hashtags alone, the risk emerges that images may be misclassified, as the presence of fashion-related hashtags does not guarantee the presence of tangible fashion items within the images themselves. We address this challenge by employing a comprehensive approach that involves not only hashtags but also image analysis techniques to ensure the accuracy and reliability of our trend predictions.

For this task, a VGG-16 (Simonyan & Zisserman, 2014) was used, as shown in Fig. 2. It was chosen because it is a fast and reliable network, that has proven to be capable to solve many classification tasks over the years. Also, after a comparative study (presented in 4.2) with other popular classification neural network has been carried out, proving the effectiveness of VGG-16 among them. The VGG-16 was trained on a novel dataset, manually created for this purpose, which consisted of both fashion and non-fashion images. Through the implementation of this CNN (Convolutional Neural Network), fashion related images have been selected. This way, all searched posts extracted with a specific hashtag that did not contain the relative fashion image were excluded.

3.2.1. Instagram image dataset

As introduced in Section 3.2, a novel dataset has been created for the purpose of classifying fashion and non-fashion images. The dataset is comprised of 700.000 images, taken from Instagram and manually selected and annotated to have a balanced dataset (350.000 fashion images and 350.000 non-fashion images). The data collection took place from January 2022 to January 2023, during which we made the decision to stratify the dataset into training, validation, and test splits on a monthly basis. This choice was motivated by the recognition that various fashion-related elements, including clothing, accessories, and people's behaviors, change in response to seasonal and weather variations. Consequently, our aim was to encompass the full spectrum of fashion trends over an entire year within the dataset. This approach serves to mitigate any temporal biases and ensures the fidelity of our model evaluation, as it closely emulates real-world situations.

² <https://github.com/instaloader/instaloader>

³ <https://www.mongodb.com/>

Table 1
Number of images and instances for each class of the dataset.

Class	Number of images	Number of instances
Backpack	4778	6123
Belt bags	3673	5313
Bucket bags	3689	6170
Clutch bags	6890	13 108
Mini bags	5239	8580
Cross-body bags	5798	9938
Shoulder bags	7096	13 524
Tote bags	6553	12 009
Pochette	3267	4878
Bowling bag	8104	15 547

3.3. Handbag detection and classification

Once all fashion images have been selected, a detection phase followed and a YOLOv5 was used for this task, as shown in Fig. 4. In particular, it was used to detect the handbags depicted inside the fashion images and classify the handbag model detected. This network was chosen as it has proven to be a good compromise between speed and accuracy in the detection domain. For this task, YOLOv5 was pre-trained on the COCO dataset (Lin et al., 2014), which is a large-scale object detection, segmentation, and captioning dataset, with more than 200.000 labeled images and 1.5 million object instances. The model was then fine-tuned on the “MADAME” (iMAGE fAshion Dataset social Media) dataset (Sciucca et al., 2022), which is a dataset consisting of 55.087 handbag models of different categories and a total of 91 380 instances. Table 1 shows the classes present in the dataset and their corresponding number of images and instances.

This phase was used to find the exact position of the bag inside the image in order to crop it and obtain a new image with the bag in foreground, ready to be analyzed.

3.4. Dominant color clustering

Analyze the current trends to support the decisions of fashion designers in the generation of new collections means to accurately understand some important visual aspect of the detected garments. The color examination of the bag is an important aspect in this scenario. A clustering by using K-means algorithm (Lloyd, 1982) was implemented to detect the dominant color of the bags. Using this unsupervised learning algorithm the pixels of the image (cropped after the bag-detection step) are grouped into a specified number of clusters based on their RGB value (projecting those pixels in a 3 dimensional space according to 3 color channels). Cluster centers are then representative of the dominant colors. In particular, 3 dominant colors have been extracted for each image and sorted by quantity of elements: the first cluster represent the background of the image, i.e. the entire area of the image that does not contain the bag that is the subject of the detection, the second and the third pertain to the 2 dominant colors of the bag. The output consists of an array of RGB triads in the range 0–255 corresponding to the 3 colors (e.g., [[255, 255, 255], [255, 25, 25], [0, 127, 127]]). Observing the problem it is clear that we deal with well-defined clusters, so the process can be sped up using the “Elkan” variation of K-Means proposed by Elkan (2003) which is more efficient on datasets with well-defined clusters by using the triangle inequality.

The integration of this information inside the JSON file allows to apply further analysis through the implementation of backend APIs to create a dominant color list, which can be obtained by time period, by geographical location or by object type.

3.5. Data analysis

As a final step, all the information collected are used to carry out different analysis through the use of backend APIs, which allow to obtain various information, such as:

- Number of handbags for each class
- Number of images with multiple handbags
- Number of images with multiple handbags of the same class
- Image classification
- Image handbag detection and classification
- Image dominant colors

These information can be very useful, as they allow to plot the most used fashion objects in a time frame (thanks to the presence of the date on the posts) and in a specific area. In fact, the localization information allows to select and analyze the trends of specific neighborhoods. As of today, fashion designers and creatives manually scout social networks, searching fashion-related posts in the most relevant cities and places; this can be a time consuming process. Social4fashion aims to automate this process, allowing for the evaluation and comparison of trends in the most fashionable cities: from Paris to New York, from Milan to London or Tokyo, with the aim of predicting future trends and supporting fashion experts in the creation of new relevant fashion collections which match the tastes of customers in a specific area.

4. Results and discussions

In this section, we present the outcomes of the extensive experiments conducted to evaluate the performance and efficacy of the machine and deep learning tasks integrated into Social4Fashion. The section is structured to provide a comprehensive understanding of the system's capabilities, beginning with a detailed overview of the experimental settings. Subsequently, we delve into the obtained results, covering diverse aspects such as classification, detection, and the detection of dominant colors. These findings collectively contribute to a comprehensive assessment of Social4Fashion's functionality and pave the way for insightful discussions regarding its strengths, limitations, and potential avenues for refinement.

4.1. Experiments settings

Fashion-non-fashion filtering is a binary classification, for this task the Instagram Images Dataset has then been split in 70% for the training set, 10% for the validation set and 20% for the test set.

Handbag detection represent an object detection task; the dataset has been split in training and test set with a 80:20 in a stratified manner in order to keep ratio between classes based on the number of object instances.

The training configuration parameters for both tasks can be found in Tables 2 and 3. Regarding the detection module, hyperparameters were chosen through empirical experimentation and by leveraging insights from previous research in the domain of object detection, utilizing the YOLOv5 architecture (Hamzenejadi & Mohseni, 2023; Mahaur et al., 2023; Yar et al., 2023). Notably, the batch size was tailored to match the GPU memory capacity and enhance the efficiency of the training process. The other configurations were kept in their default state, aligned with the original YOLOv5 model (Jocher, 2020). The same approach has been taken in regards to the classification model (VGG16), going through empirical experimentations and leveraging insights from similar works that deal with the classification task by using VGG16 (Ilias et al., 2023; Nigam et al., 2021; Wang et al., 2022). Other parameters were kept in their default state, in line with the original VGG16 model (Simonyan & Zisserman, 2014).

We employed an early stopping mechanism for both the classification and detection tasks. This mechanism was based on monitoring the loss value during training for both classification and detection. Specifically, the early stopping condition was utilized to track improvements in validation loss with a patience of 10 epochs. To maintain training efficiency, we set this evaluation to occur every 10 epochs. The best model is preserved following each evaluation. The model selected for fashion/non-fashion classification is the VGG16, which has an average

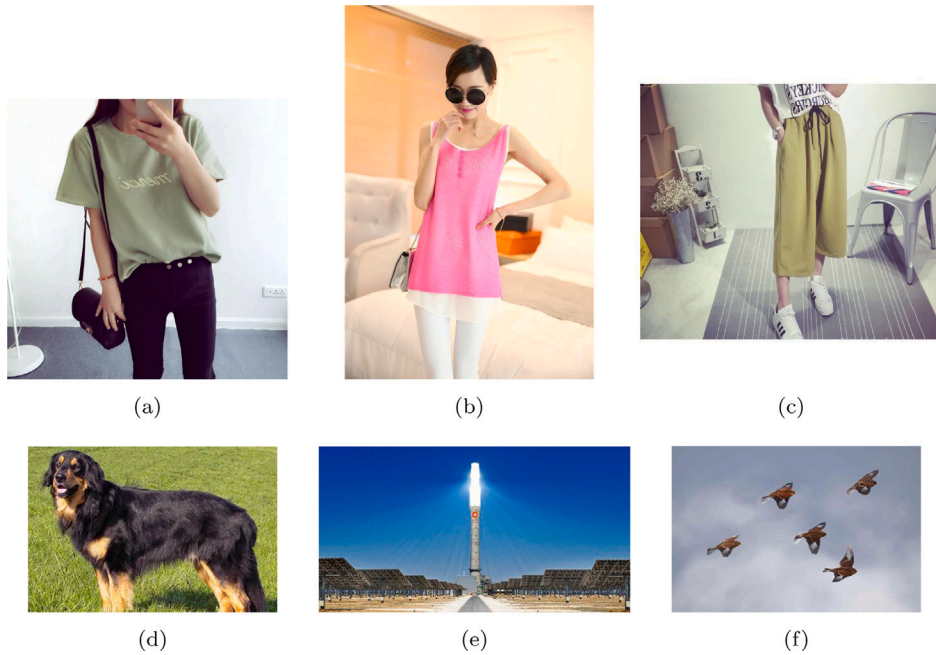


Fig. 3. Examples of fashion and non-fashion images. Images (a) (b) (c) are examples of fashion images. Images (d) (e) (f) are examples of non-fashion images.

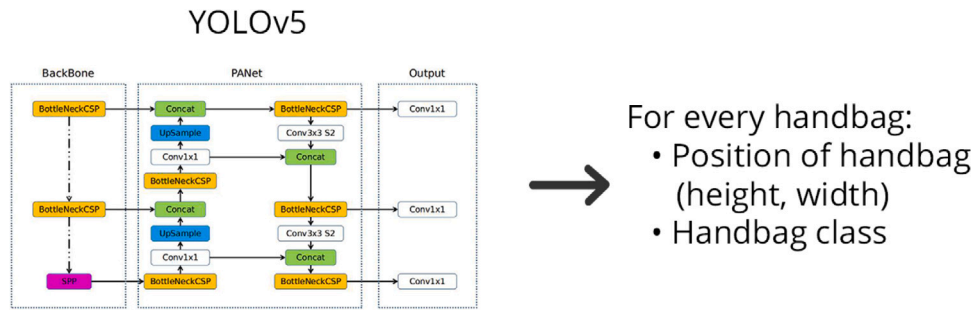


Fig. 4. Workflow of the handbags detection and classification phase with YOLOv5.

Table 2
Configuration values for training of classification model.

Parameter	Value
Learning Rate	0.00001
Batch size	32
Optimizer	Adam
Epochs	50

Table 3
Configuration values for training of detection model.

Parameter	Value
Learning Rate	0.01
Batch size	128
Optimizer	SDG
Momentum	0.94
Epochs	150

inference time of 30 ms based on our experiments on a NVIDIA T4 GPU. Training time on the same architecture took 7 h. The model chosen for bag detection is YOLOv5s, which is the compact version of YOLOv5. Its training duration on an NVIDIA T4 GPU amounted to 9 h. Inference time registers at 8 ms on the same architecture.

4.2. Classification results

The first step involved the binary classification of images in fashion and non-fashion, using the novel dataset created from Instagram images. As stated before, VGG-16 network has been chosen; this choice was made after comparing 3 other popular networks used for image classification, namely ResNet50, DenseNet121 and EfficientNet. Table 4 shows the quantitative results of the comparison in terms of Accuracy, Precision, Recall and F1-score on the test set after 50 epochs. Looking at the metrics results, it can be seen that VGG-16 is the best performing network, proving its effectiveness and efficiency. Confidence intervals, which play a crucial role in the estimation of unknown parameters, have been used as a measure of variability (Steinbach et al., 2022). Assumptions must be made about the distribution of this accuracy value when only one estimate is available, such as the accuracy from a single test set. It is often assumed that the accuracies calculated from different samples are normally distributed. In an ideal situation, we would have access to the distribution of the sample from our test set. In this scenario, it is possible to examine the range of values within which it is expected that 95% of the accuracy values will fall. While this approach is desirable, it is not practical because of the constraint of having a finite number of test sets. A practical solution to this limitation is bootstrapping, a method of estimating the sampling distribution. This provides a valuable means of approximating the



Fig. 5. Examples of bag detection with GRAD-CAM for model interpretability.

Table 4
Classification results (95% CI).

CNNs	Accuracy	Precision	Recall	F1-score
ResNet 50	0.91 (0.87–0.94)	0.92 (0.85–0.95)	0.92 (0.86–0.93)	0.92 (0.85–0.94)
VGG16	0.97 (0.95–1.00)	0.94 (0.92–0.96)	0.94 (0.92–0.97)	0.94 (0.92–0.96)
DenseNet 121	0.89 (0.83–0.94)	0.88 (0.81–0.93)	0.91 (0.85–0.94)	0.89 (0.83–0.93)
EfficientNet	0.93 (0.85–0.96)	0.93 (0.85–0.95)	0.95 (0.89–1.00)	0.94 (0.87–0.97)

distribution of accuracy values by generating multiple samples with replacement from a single random sample. We chose to implement bootstrapping specifically on the test set, where we maintain a fixed model and proceed to resample the test set with replacement. This approach avoids the need to retrain the model, which is often impractical in deep learning approaches, and is particularly attractive in deep learning scenarios. Following best practices in the field, we chose 200 bootstrap rounds (Efron & Tibshirani, 1994). For each round, the metrics were calculated and then averaged to give the single value. In contrast, the 2.5th and 97.5th percentiles of the bootstrap sample distribution (metric values) were taken as our upper and lower confidence limits. Some qualitative results are also provided in Fig. 3.

4.3. Detection results

The second experiment involved the handbag detection and classification phase using the popular object detector YOLOv5. Results in terms of Precision, Recall and Mean Average Precision on the test set have been reported in Table 5. As it can be seen, results are pretty satisfactory, with a total average Precision of 0.842, Recall of 0.741 and mAP@.5 of 0.787; there are however some classes that the network struggle to detect and classify (e.g., backpacks), while other types of handbags are detected and classified with a much higher rate (e.g., clutch bags, shoulder bags, bowling bags); these behaviors can be attributed to different factors, such as the different number of elements for each class in the MADAME dataset (which the network was fine-tuned on), as the classes with more images show better performance metrics, while classes with less images do not perform as good as

the others; another reason could be attributed to the fact that some handbags can fit into more than one class (e.g., a clutch bag could also be a mini bag) so, in some cases, a clear distinction between the classes cannot be made; this can affect the accuracy of the network. Adhering to the same principles applied during the classification phase, we employ a methodology to assess metric variability by introducing 95% confidence intervals. These intervals are derived from 200 bootstrap rounds conducted on the test set. Some qualitative results are also provided in Fig. 6.

To account for the class imbalance, although we consider it not severe (the maximum ratio between classes for the number of instances is approximately 3:1, between the 'Bowling bag' class with 15547 instances and the 'Pochette' class with 4878 instances) we used a Precision-Recall metric analysis. Following procedures reported in literature (Padilla et al., 2021), we have used the PASCAL VOC metric of mAP, which is widely recognized and adopted for evaluating object detection models (Everingham et al., 2010). mAP (mean Average Precision) is designed to address the presence of imbalanced classes, since it considers precision and recall. The decision to employ well-established metrics like PASCAL VOC metrics underscores our commitment to robust evaluations conducted in our study, even in scenarios involving class imbalance (for this reason the accuracy has not been taken into consideration). Model interpretability has been investigated using Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2017), designed to provide a visual explanation of which parts of the image contribute most significantly to the model's prediction. The resulting heatmap employs a range of colors to indicate the level of importance associated with the classification output. Darker shades in the heatmap correspond to regions that exert substantial influence on the classification result, while lighter shades suggest a relatively weaker impact from those specific areas. By emphasizing these particular regions, valuable insights can be gained into the criteria that the models emphasize when making assessments about prevailing trends. In Fig. 5 some examples are provided and they reveals that the bag closure plays an important role in the classification. For instance, the examination illuminates the noteworthy role of zippered closures in the context of Backpacks (Fig. 5(a)). The presence of zippers in Backpacks stands out as a defining characteristic that aids in accurately categorizing them. Likewise, the analysis delves into the significance of buckle closures, particularly in the realm of Clutch Bags (Fig. 5(c)). Buckle closures emerge as a key feature that significantly contributes to the successful

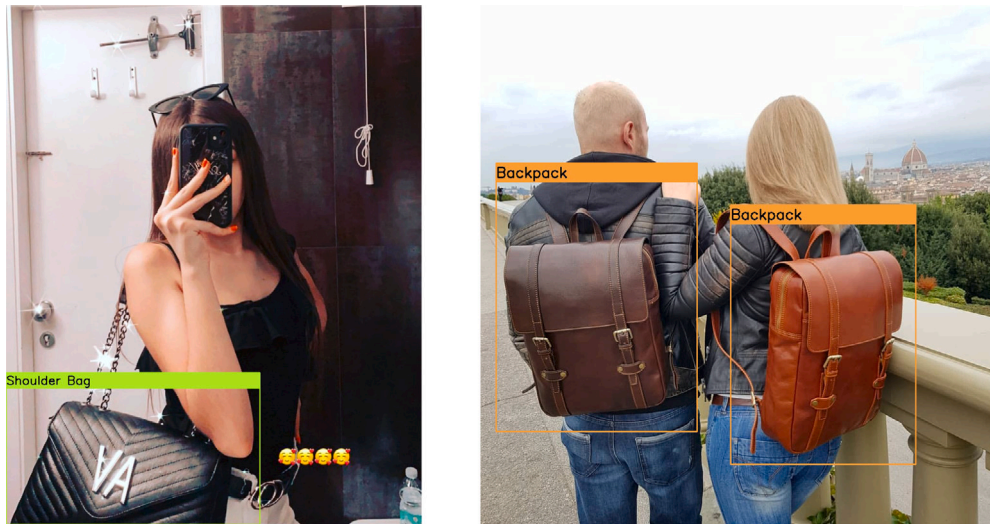


Fig. 6. Examples of bag detection and classification.

Table 5

Results metrics of detection phase on test set (95% CI).

Class	Precision	Recall	mAP@.5
Backpack	0.702 (0.651–0.733)	0.714 (0.647–0.740)	0.714 (0.670–0.751)
Belt bags	0.838 (0.80–0.859)	0.724 (0.632–0.754)	0.737 (0.69–0.774)
Bucket bags	0.821 (0.751–0.878)	0.809 (0.771–0.852)	0.754 (0.701–0.79)
Clutch bags	0.912 (0.877–0.945)	0.813 (0.787–0.871)	0.827 (0.78–0.867)
Mini bags	0.837 (0.813–0.899)	0.754 (0.703–0.811)	0.829 (0.791–0.879)
Cross-body bags	0.856 (0.812–0.897)	0.757 (0.706–0.799)	0.819 (0.777–0.866)
Shoulder bags	0.901 (0.878–0.957)	0.764 (0.714–0.798)	0.757 (0.721–0.821)
Tote bags	0.863 (0.815–0.893)	0.747 (0.711–0.792)	0.815 (0.754–0.832)
Pochette	0.865 (0.831–0.917)	0.814 (0.751–0.856)	0.820 (0.779–0.852)
Bowling bag	0.946 (0.913–0.972)	0.848 (0.812–0.885)	0.855 (0.824–0.904)
Total	0.832 (0.807–0.873)	0.754 (0.711–0.789)	0.777 (0.732–0.825)

classification of Clutch Bags. Also in the case of Shoulder Bags, the types of closures, such as zips or magnetic clasps, are highlighted as pivotal elements affecting the precision of categorizing Shoulder Bags (Fig. 5(d)). Furthermore, the study elucidates the role of closure designs in Belt Bags (Fig. 5(c)). The manner in which these bags are secured, whether through buckle closures or other distinctive fastening methods, emerges as a fundamental aspect shaping the classification results.

4.4. Dominant color detection

Qualitative results for dominant color detection are shown in Fig. 7. As it can be seen, after cropping the image around the bounding box of the detected handbag, the 3 main colors are extracted. A note can be made regarding the resulting colors: due to the nature of the rectangular selection made by the detection, some foreign elements can happen to be included; while, in most cases, this does not influence the resulting RGB triplets, in some instances colors not relative to the

handbag can be captured (e.g., in the referenced figure: the person's arm, the background). This could be solved by applying semantic segmentation on the handbag instead of object detection.

5. Social4Fashion data visualization

Dashboards present data in an easily digestible format, making it simpler for users to understand and draw insights from the data. They allow real-time monitoring of key performance indicators and metrics, providing visibility into business performances and opportunities for improvement. By providing relevant and up-to-date information, they enable informed decision making and timely action. They can be customized to display specific data and Key Performance Indicators (KPIs), providing a tailored view of relevant information. They also support collaboration through the sharing of dashboards among team members. Our framework incorporates a user-friendly dashboard designed to provide a comprehensive and interactive environment for fashion professionals and researchers to explore trends, patterns, and insights derived from our deep learning-based analysis. The dashboards serve as a crucial bridge between the backend processes and the end users seeking actionable insights. Our user-friendly dashboard serves as a dynamic window into the intricate landscape of fashion trends. The design is thought to make it as understandable and intuitive as possible. Engineered to be used by both technical and non-technical users, the dashboard provides an intuitive interface that encapsulates the essence of fashion analysis. Through simplified trend charts, users can seamlessly comprehend the temporal evolution of fashion-related content. The dashboard's interactive features empower users to zoom in on specific time periods, apply filters, and explore color clusters, ensuring an engaging and informative experience. Social4Fashion dashboards are meticulously designed using the powerful capabilities of PowerBI, a leading business intelligence tool.⁴ Leveraging intuitive interface and robust features, these dashboards offer a seamless user experience for exploring intricate social media data related to handbags. The design revolves around user-friendliness and interactivity, with a focus on delivering actionable insights at a glance. The layout is thoughtfully organized, presenting a logical flow from top to bottom, allowing users to effortlessly interact with the various elements. The color-coded handbag type filter and geolocation map make navigation intuitive, while the time filter enables users to seamlessly zoom into specific time periods. The visualizations, including the dominant colors

⁴ <https://powerbi.microsoft.com/>

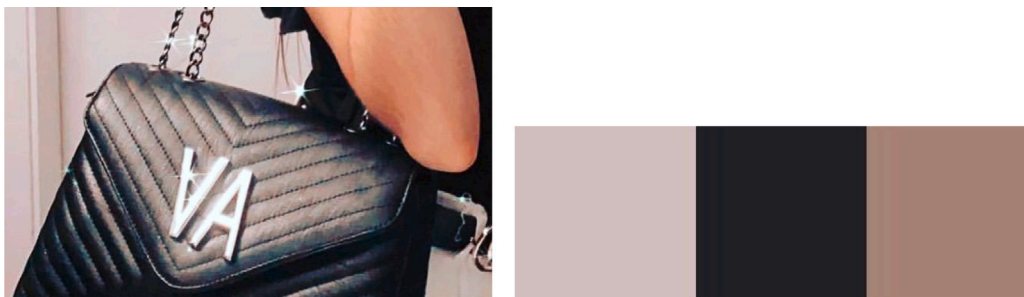
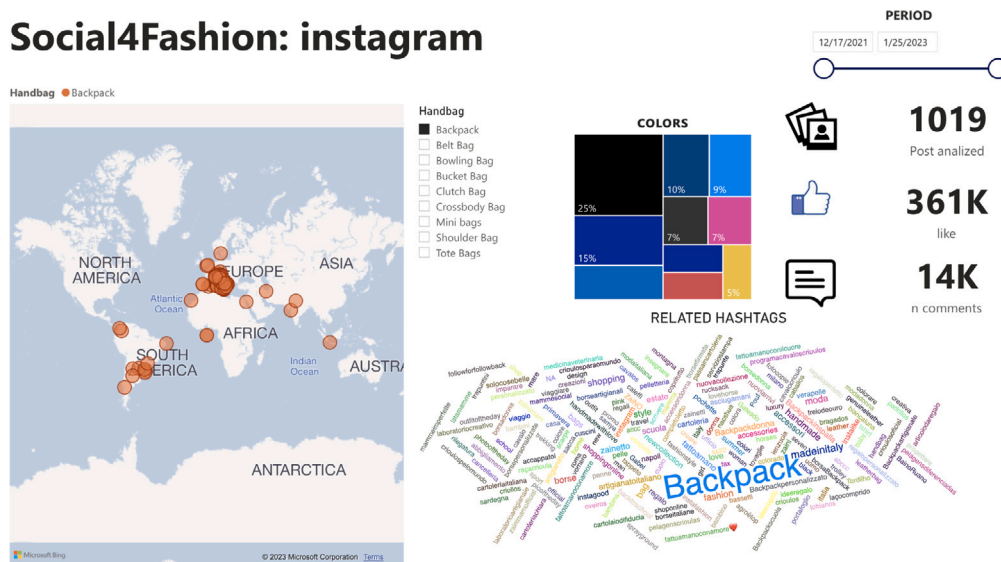
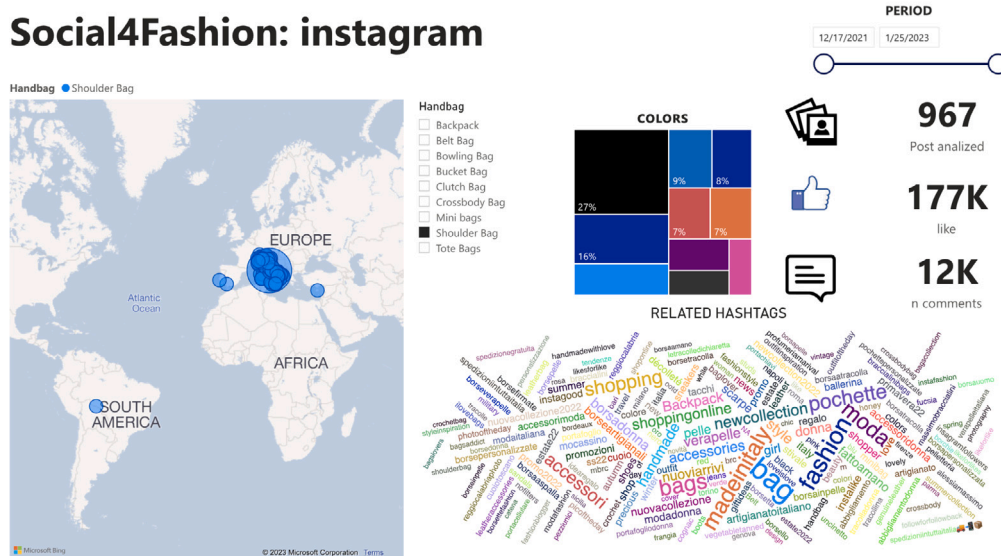


Fig. 7. Examples of color dominance detection, detected colors on the right: $[[81,74,74], [22,25,40], [41,27,23]]$.



(a) The figure shows the filtered data for the Backpack class.



(b) The figure shows the filtered data for the Shoulder bag class.

Fig. 8. Examples of dashboards resulting from the previous experiments.

grid and related hashtags section, are crafted to be visually engaging, facilitating quick comprehension of trends. The KPIs are prominently displayed, ensuring that essential metrics are front and center. Overall, the Social4Fashion Dashboards harmonize aesthetic appeal with functionality, empowering users to explore, analyze, and extract meaningful information from the world of handbag-related social media data.

Fig. 8 shows some examples of a dashboard that can be created with data extracted by the system, with different filters applied. The dashboard allows to control the status of the extracted data and, moreover, it allows users to evaluate the trends of posts relating to the various categories of handbags in a specific time and location.

Following the result examples in previous sections, we examined the data collected for classes ‘Backpack’ and ‘Shoulder bag’. The dashboard allows users to monitor key performance indicators and metrics in real time. In our case, filtering by classes ‘Backpack’ and ‘Shoulder bag’ we can quickly see the number of posts analyzed, the level of interaction, related hashtags and their frequency, the frequency of the colors of the bags analyzed, the geographical areas from where the posts are published based on the selected period.

A more detailed explanation of the different elements that compose the given dashboards follows:

- Handbag type filter: a filter that allows to narrow the analysis by selection only some types of handbags,
- Time filter: a filter that allows to select a specific time frame,
- Searched hashtags by geolocation: a map is present, highlighting the location where different handbags are present the most; colors are representative of the different handbag types. By selecting a specific area, a filter is automatically applied on the whole dashboard, showing data only from that specific area,
- Dominant colors grid: a grid showing the colors resulting from the analysis of dominant colors,
- Related hashtags: hashtags most present in the analyzed posts,
- 3 KPIs: number of posts taken into account, number of likes in the analyzed posts, number of comments in the analyzed posts.

Date and location filters are applied on every element present in the dashboard, allowing industry experts to evaluate trends by location, period, type of handbag and color. This can be very useful to domain experts to evaluate any deviation from the expected trends in real time.

6. Conclusions and future works

The attention to fashion has grown dramatically with the development of social networks. Trends are defined very quickly online, by both influencers and people in general; every day thousands of fashion images are posted and creatives are forced to manually go through them in order to better understand new trends and get inspired; to automate this whole process, Social4Fashion has been created: it is an expert system which objective is to extract and analyze images and data from social networks, in order to evaluate current trends and predict future ones, in specific geographical areas, providing support to fashion experts in the creation of future collections. The system proposed in this work focused on the creation of a pipeline which, starting from Instagram images and data, performed different consecutive tasks in order to extract useful information; the tasks consisted of the classification of images in fashion and non-fashion (in order to remove non relevant elements), the detection of handbags in the images (and the subsequent classification of the handbag type) and clustering to extract the dominant colors from the image. All these information have been used to create dashboards and visualizations to aid creatives in viewing trends.

We foresee a future direction for the Social4Fashion framework that embraces a more inclusive approach to trend prediction. Building on the recognition that trends often originate from non-fashion domains, we plan to refine our system to incorporate a broader spectrum of influences, such as architectural aesthetics, cultural shifts, and geographical

inspirations. This evolution will involve developing methodologies to gather and analyze data from diverse sources, thereby enhancing the framework’s capacity to predict trends that emerge from a rich tapestry of factors. Additionally, other possible future developments could see the application of the same system to other accessories or garments by performing searches with different hashtags (e.g., types of shoes, t-shirts, or pants). We also foresee the potential application of semantic segmentation, as a substitute for object detection, to refine the process of dominant color clustering. This enhancement would enable the removal of background elements entirely, thereby improving the accuracy of color analysis. In recognition of the pivotal role that social media platforms hold in discerning current fashion trends and the expansion of both platforms and their user bases, our approach’s modular nature lends itself to effortless adaptation across diverse social media ecosystems. Our initial selection of Instagram was grounded in its status as a hub for fashion-related content, including dedicated pages from fashion brands, luxury labels, and influencers. However, our future directions involve extending our methodology to encompass a wider array of social media platforms, each involving to distinct demographics and user behaviors. This expansion will culminate in an in-depth comprehension of fashion trends that transcends the constraints of individual platforms. Through these strategic approach, we aspire to improve Social4Fashion as framework that captures the multifaceted essence of fashion while enhancing predictive accuracy through a diverse spectrum of data sources and fashion items. Finally, the novel dataset created through this work provides a robust foundation for future research endeavors in this domain. It holds the promise of facilitating subsequent investigations, training initiatives, and the development of innovative methodologies in the domain of fashion trend analysis.

CRedit authorship contribution statement

Emanuele Balloni: Conceptualization, Methodology, Software. **Rocco Pietrini:** Software, Validation. **Matteo Fabiani:** Visualization, Investigation. **Emanuele Frontoni:** Data curation, Writing – review & editing. **Adriano Mancini:** Software. **Marina Paolanti:** Conceptualization, Methodology, Software, Supervision, Writing – original draft.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Marina Paolanti reports was provided by University of Macerata. Marina Paolanti reports a relationship with University of Macerata that includes: Marina Paolanti has patent pending to No patent. Nothing to declare.

Data availability

Data will be made available on request.

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References

- Al-Halah, Z., Stiefelwagen, R., & Grauman, K. (2017). Fashion forward: Forecasting visual style in fashion. <http://dx.doi.org/10.48550/ARXIV.1705.06394>, URL: <https://arxiv.org/abs/1705.06394>.
- Balim, C., & Özkan, K. (2023). Diagnosing fashion outfit compatibility with deep learning techniques. *Expert Systems with Applications*, 215, Article 119305. <http://dx.doi.org/10.1016/j.eswa.2022.119305>, URL: <https://www.sciencedirect.com/science/article/pii/S0957417422023235>.
- Cho, S. (2019). Creative design thinking process: Fashion is cyclical then, now, and future. In *International textile and apparel association annual conference proceedings*, vol. 76, no. 1. Iowa State University Digital Press.
- Cucurull, G., Taslakian, P., & Vazquez, D. (2019). Context-aware visual compatibility prediction. <http://dx.doi.org/10.48550/ARXIV.1902.03646>, URL: <https://arxiv.org/abs/1902.03646>.
- Della Sciacca, L., Balloni, E., Mameli, M., Frontoni, E., Zingaretti, P., & Paolanti, M. (2022). StyleTrendGAN: A deep learning generative framework for fashion bag generation. In *Image analysis and processing. ICIAP 2022 workshops: ICIAP international workshops* (pp. 191–202). Berlin, Heidelberg: Springer-Verlag. http://dx.doi.org/10.1007/978-3-031-13324-4_17.
- Ding, Y., Ma, Y., Liao, L., Wong, W. K., & Chua, T. S. (2022). Leveraging multiple relations for fashion trend forecasting based on social media. *IEEE Transactions on Multimedia*, 24, 2287–2299. <http://dx.doi.org/10.1109/TMM.2021.3078907>.
- Efron, B., & Tibshirani, R. J. (1994). *An introduction to the bootstrap*. CRC Press.
- Elkan, C. (2003). Using the triangle inequality to accelerate k-means. In *Proceedings of the 20th international conference on machine learning* (pp. 147–153).
- Everingham, M., Van Gool, L., Williams, C. K., Winn, J., & Zisserman, A. (2010). The pascal visual object classes (voc) challenge. *International Journal of Computer Vision*, 88, 303–338.
- Hamzenejadi, M. H., & Mohseni, H. (2023). Fine-tuned YOLOv5 for real-time vehicle detection in UAV imagery: Architectural improvements and performance boost. *Expert Systems with Applications*, 231, Article 120845. <http://dx.doi.org/10.1016/j.eswa.2023.120845>, URL: <https://www.sciencedirect.com/science/article/pii/S0957417423013477>.
- Han, A., Kim, J., & Ahn, J. (2022). Color trend analysis using machine learning with fashion collection images. *Clothing and Textiles Research Journal*, 40(4), 308–324. <http://dx.doi.org/10.1177/0887302X21995948>.
- Ilias, L., Askounis, D., & Psarras, J. (2023). Multimodal detection of epilepsy with deep neural networks. *Expert Systems with Applications*, 213, Article 119010. <http://dx.doi.org/10.1016/j.eswa.2022.119010>, URL: <https://www.sciencedirect.com/science/article/pii/S0957417422020280>.
- Iwata, T., Watanabe, S., & Sawada, H. (2011). Fashion coordinates recommender system using photographs from fashion magazines. In *Proceedings of the twenty-second international joint conference on artificial intelligence - volume volume three* (pp. 2262–2267). AAAI Press.
- Jain, A. K., & Dubes, R. C. (1988). *Algorithms for clustering data*. Prentice-Hall, Inc.
- Jain, S., & Kumar, V. (2020). Garment categorization using data mining techniques. *Symmetry*, 12(6), <http://dx.doi.org/10.3390/sym12060984>, URL: <https://www.mdpi.com/2073-8994/12/6/984>.
- Jocher, G. (2020). YOLOv5 by ultralytics. <http://dx.doi.org/10.5281/zenodo.3908559>, URL: <https://github.com/ultralytics/yolov5>.
- Kang, W. C., Kim, E., Leskovec, J., Rosenberg, C., & McAuley, J. (2018). Complete the look: Scene-based complementary product recommendation. <http://dx.doi.org/10.48550/ARXIV.1812.01748>, URL: <https://arxiv.org/abs/1812.01748>.
- Kiapour, M. H., Han, X., Lazebnik, S., Berg, A. C., & Berg, T. L. (2015). Where to buy it: Matching street clothing photos in online shops. In *2015 IEEE international conference on computer vision* (pp. 3343–3351). <http://dx.doi.org/10.1109/ICCV.2015.382>.
- Kim, H. J., Lee, D. H., Niaz, A., Kim, C. Y., Memon, A. A., & Choi, K. N. (2021). Multiple-clothing detection and fashion landmark estimation using a single-stage detector. *IEEE Access*, 9, 11694–11704. <http://dx.doi.org/10.1109/ACCESS.2021.3051424>.
- Kolisnik, B., Hogan, I., & Zulkernine, F. (2021). Condition-CNN: A hierarchical multi-label fashion image classification model. *Expert Systems with Applications*, 182, Article 115195. <http://dx.doi.org/10.1016/j.eswa.2021.115195>, URL: <https://www.sciencedirect.com/science/article/pii/S0957417421006291>.
- Lai, P., & Westland, S. (2020). Machine learning for colour palette extraction from fashion runway images. *International Journal of Fashion Design, Technology and Education*, 13(3), 334–340. <http://dx.doi.org/10.1080/17543266.2020.1799080>.
- Li, Y., Cao, L., Zhu, J., & Luo, J. (2017). Mining fashion outfit composition using an end-to-end deep learning approach on set data. *IEEE Transactions on Multimedia*, 19(8), 1946–1955. <http://dx.doi.org/10.1109/TMM.2017.2690144>.
- Li, K., Chong, M. J., Zhang, J., & Liu, J. (2021). Toward accurate and realistic outfits visualization with attention to details. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 15546–15555).
- Lin, T. Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft coco: Common objects in context. In *European conference on computer vision* (pp. 740–755). Springer.
- Lin, Y., Tran, S. D., & Davis, L. S. (2019). Fashion outfit complementary item retrieval. CoRR abs/1912.08967. [arXiv:1912.08967](https://arxiv.org/abs/1912.08967).
- Lloyd, S. P. (1982). Least squares quantization in PCM. *Institute of Electrical and Electronics Engineers. Transactions on Information Theory*, 28, 129–136.
- Lu, Z., Hu, Y., Chen, Y., & Zeng, B. (2021). Personalized outfit recommendation with learnable anchors. In *2021 IEEE/CVF conference on computer vision and pattern recognition* (pp. 12717–12726). <http://dx.doi.org/10.1109/CVPR46437.2021.01253>.
- Lu, Z., Hu, Y., Jiang, Y., Chen, Y., & Zeng, B. (2019). Learning binary code for personalized fashion recommendation. In *2019 IEEE/CVF conference on computer vision and pattern recognition* (pp. 10554–10562). <http://dx.doi.org/10.1109/CVPR.2019.01081>.
- Ma, Y., Ding, Y., Yang, X., Liao, L., Wong, W. K., & Chua, T. S. (2020). Knowledge enhanced neural fashion trend forecasting. <http://dx.doi.org/10.48550/ARXIV.2005.03297>, URL: <https://arxiv.org/abs/2005.03297>.
- Ma, Y., Yang, X., Liao, L., Cao, Y., & Chua, T. S. (2019). Who, where, and what to wear? extracting fashion knowledge from social media. In *Proceedings of the 27th ACM international conference on multimedia* (pp. 257–265).
- Mahaur, B., Mishra, K., & Kumar, A. (2023). An improved lightweight small object detection framework applied to real-time autonomous driving. *Expert Systems with Applications*, 234, Article 121036. <http://dx.doi.org/10.1016/j.eswa.2023.121036>, URL: <https://www.sciencedirect.com/science/article/pii/S0957417423015385>.
- Mall, U., Matzen, K., Hariharan, B., Snavely, N., & Bala, K. (2019). GeoStyle: Discovering fashion trends and events. <http://dx.doi.org/10.48550/ARXIV.1908.11412>, URL: <https://arxiv.org/abs/1908.11412>.
- Mameli, M., Paolanti, M., Pietrini, R., Pazzaglia, G., Frontoni, E., & Zingaretti, P. (2021). Deep learning approaches for fashion knowledge extraction from social media: a review. *IEEE Access*.
- Nigam, B., Nigam, A., Jain, R., Dodia, S., Arora, N., & Annappa, B. (2021). COVID-19: Automatic detection from X-ray images by utilizing deep learning methods. *Expert Systems with Applications*, 176, Article 114883. <http://dx.doi.org/10.1016/j.eswa.2021.114883>, URL: <https://www.sciencedirect.com/science/article/pii/S0957417421003249>.
- Padilla, R., Passos, W. L., Dias, T. L., Netto, S. L., & Da Silva, E. A. (2021). A comparative analysis of object detection metrics with a companion open-source toolkit. *Electronics*, 10(3), 279.
- Rostamzadeh, N., Hosseini, S., Boquet, T., Stokowiec, W., Zhang, Y., Jauvin, C., & Pal, C. (2018). Fashion-gen: The generative fashion dataset and challenge. <http://dx.doi.org/10.48550/ARXIV.1806.08317>, URL: <https://arxiv.org/abs/1806.08317>.
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision* (pp. 618–626).
- Seo, Y., & shik Shin, K. (2019). Hierarchical convolutional neural networks for fashion image classification. *Expert Systems with Applications*, 116, 328–339. <http://dx.doi.org/10.1016/j.eswa.2018.09.022>, URL: <https://www.sciencedirect.com/science/article/pii/S0957417418305992>.
- Shimizu, R., Saito, Y., Matsutani, M., & Goto, M. (2023). Fashion intelligence system: An outfit interpretation utilizing images and rich abstract tags. *Expert Systems with Applications*, 213, Article 119167. <http://dx.doi.org/10.1016/j.eswa.2022.119167>, URL: <https://www.sciencedirect.com/science/article/pii/S0957417422021856>.
- Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. <http://dx.doi.org/10.48550/ARXIV.1409.1556>, URL: <https://arxiv.org/abs/1409.1556>.
- Song, X., Feng, F., Liu, J., Li, Z., Nie, L., & Ma, J. (2017). Neurostylist: Neural compatibility modeling for clothing matching. In *Proceedings of the 25th ACM international conference on multimedia* (pp. 753–761).
- Steinbach, P., Gernhardt, F., Tanveer, M., Schmerler, S., & Starke, S. (2022). Machine learning state-of-the-art with uncertainties. [arXiv preprint arXiv:2204.05173](https://arxiv.org/abs/2204.05173).
- Veit, A., Kovacs, B., Bell, S., McAuley, J. J., Bala, K., & Belongie, S. J. (2015). Learning visual clothing style with heterogeneous dyadic co-occurrences. CoRR abs/1509.07473. [arXiv:1509.07473](https://arxiv.org/abs/1509.07473).
- Wang, W., Xu, Y., Shen, J., & Zhu, S. C. (2018). Attentive fashion grammar network for fashion landmark detection and clothing category classification. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4271–4280).
- Wang, Y., Zhang, W., Huang, D., Liu, Y., & Zhu, J. (2022). Multi-scale features fused network with multi-level supervised path for crowd counting. *Expert Systems with Applications*, 200, Article 116949. <http://dx.doi.org/10.1016/j.eswa.2022.116949>, URL: <https://www.sciencedirect.com/science/article/pii/S0957417422003785>.
- Wazarkar, S., & Keshavamurthy, B. N. (2020). Social image mining for fashion analysis and forecasting. *Applied Soft Computing*, 95, Article 106517.
- Wu, H., Gao, Y., Guo, X., Al-Halah, Z., Rennie, S., Grauman, K., & Feris, R. (2019). Fashion IQ: A new dataset towards retrieving images by natural language feedback. <http://dx.doi.org/10.48550/ARXIV.1905.12794>, URL: <https://arxiv.org/abs/1905.12794>.
- Yar, H., Khan, Z. A., Ullah, F. U. M., Ullah, W., & Baik, S. W. (2023). A modified YOLOv5 architecture for efficient fire detection in smart cities. *Expert Systems with Applications*, 231, Article 120465. <http://dx.doi.org/10.1016/j.eswa.2023.120465>, URL: <https://www.sciencedirect.com/science/article/pii/S0957417423009673>.
- Yu, Y., Hui, C. L., & Choi, T. M. (2012). An empirical study of intelligent expert systems on forecasting of fashion color trend. *Expert Systems with Applications*, 39(4), 4383–4389. <http://dx.doi.org/10.1016/j.eswa.2011.09.153>, URL: <https://www.sciencedirect.com/science/article/pii/S095741741101476X>.
- Zhou, S., Barnes, L., McCormick, H., & Cano, M. B. (2021). Social media influencers' narrative strategies to create eWOM: A theoretical contribution. *International Journal of Information Management*, 59, Article 102293.